Causal Reasoning P-Quiz (Short)

- **Quiz Type**: Practice Quiz
- **Total Points**: 16
- **Questions**: 16
- **Due Date**: May 10, 2026

Question 1 (1 pt)

If intervening on X changes Y, but intervening on Y does not change X, then: [✓] A. X causes Y [ ] B. X and Y are independent [ ] C. Y causes X [ ] D. A common cause exists

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Question 2 (1 pt)

In a system with variables X, Y, Z → W, a valid causal assignment must: [ ] A. Specify all variables [ ] B. Specify only effect variables [✓] C. Specify all non-effect variables [ ] D. Specify only one variable

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Question 3 (1 pt)

A valid test pair for X must: [ ] A. Change X and at least one other variable [✓] B. Change X while keeping all other variables fixed [ ] C. Keep X fixed [ ] D. Change all variables

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Question 4 (1 pt)

X is a direct cause of Y if: [ ] A. X and Y are correlated [ ] B. X always determines Y [ ] C. X precedes Y [✓] D. There exists at least one test pair where changing X changes Y

Question 5 (1 pt)

If X → Z → Y, then: [ ] A. X is not a cause of Y [ ] B. X is a direct cause of Y [ ] C. X and Y are independent [✓] D. X is an indirect cause of Y

Question 6 (1 pt)

A system where identical assignments produce different outcomes is: [ ] A. Deterministic [✓] B. Indeterministic or missing variables [ ] C. Independent [ ] D. Non-causal

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Question 7 (1 pt)

In X → Y ← Z: [ ] A. X causes Z [ ] B. Z causes X [✓] C. X and Z are independent unless conditioning on Y (Collider / V-structure) [ ] D. X and Z are always dependent

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Question 8 (1 pt)

A study finds: - *Social Media (S) ↑ → Performance (P) ↓ - Stress (T) ↑ → S ↑ and P ↓ - Sleep (L) is reduced by S and improves P (Note: first relation is read as, increase in social media, increases/decreases performance).* Which is MOST accurate? [✓] A. T is a confounder and L is a mediator [ ] B. S directly causes P [ ] C. S has only indirect effects on P [ ] D. No causal relationship exists

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Question 9 (1 pt)

You observe: - *Students who study more perform better - But when students are forced to study more: Some improve, others worsen* Which is MOST accurate? [ ] A. Study has no causal effect [✓] B. The system is indeterministic or heterogeneous [ ] C. Observation was incorrect [ ] D. Reverse causation exists

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Question 10 (1 pt)

Exercise (E) correlates with Mental Health (M). Motivation (K) affects both E and M. Which is MOST accurate? [✓] A. K is a common cause (confounder) [ ] B. E causes M [ ] C. M causes E [ ] D. No relationship exists

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Question 11 (1 pt)

System: - *A → B → D - A → C → D* Which is TRUE? [ ] A. A directly causes D [✓] B. A only indirectly affects D [ ] C. B and C are independent of A [ ] D. D causes A

Question 12 (1 pt)

A drug increases recovery rates in observational data. In trials, effect disappears. Which is MOST likely? [ ] A. Drug is ineffective [ ] B. Trial is flawed [ ] C. Reverse causation [✓] D. Observational data had confounding

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Question 13 (1 pt)

Two patients receive identical treatment but have different outcomes. Which is the MOST accurate? [ ] A. No causation exists [ ] B. System is deterministic [ ] C. Measurement error [✓] D. System is indeterministic or heterogeneous

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Question 14 (1 pt)

Job Skill (X) → Hiring (Y) ← Luck (Z). If we analyze only hired individuals (conditioning on collider Y): [✓] A. X and Z become dependent (Collider bias / Berkson's bias) [ ] B. X and Z become independent [ ] C. X causes Z [ ] D. Z causes X

Question 15 (1 pt)

System has the following response: X → Z → Y. If Z is unobserved: [✓] A. X appears to directly cause Y [ ] B. X appears unrelated to Y [ ] C. Y causes X [ ] D. No inference possible

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Question 16 (1 pt)

Observed: X and Y correlated; X and Z correlated; Y and Z independent. Which structure is MOST plausible? [ ] A. X → Y → Z [ ] B. X ← Y → Z [ ] C. X → Y ← Z [✓] D. Y → X → Z (or Y ← X → Z where X is the common cause/intermediate between independent Y and Z)